

Commissioned Review:

Gammaproteobacterial RNase E-Centered RNA Degradation in *Pseudomonas putida* KT2440

Target taxon: *Pseudomonas putida* KT2440 (PSEPK; NCBI taxon 160488; proteome UP000000556) **Target bucket:** KEGG `ppu03018` "RNA degradation" (15 primary genes; module area `other_kegg_pathway`) **Module:** Reusable RNase E-initiated decay module — endonucleolytic initiation → optional DEAD-box unwinding → processive 3'→5' exonucleolysis, with RhlB/RhlE and PNPase/RNase R as lineage-variable implementations.

1. Executive Summary

The RNase E-centered RNA-degradation module encoded by KEGG bucket `ppu03018` is **satisfiable in *Pseudomonas putida* KT2440**. Every mechanistic step of the working module can be assigned to at least one candidate gene, and — importantly — **both OR-branches of the two variant steps are encoded in the genome**. Step 1 (endonucleolytic initiation and degradosome scaffolding) is covered by *rne*/RNase E (PP_1905). Step 2 (ATP-dependent unwinding of structured RNA) is covered by RhlB (PP_1295) with additional DEAD-box paralogs RhlE (PP_4980), RhlE-I/RhpA (PP_4766), and the cold-shock helicase DeaD/CsdA (PP_1868). Step 3 (processive 3'→5' exonucleolysis) is covered by **both** PNPase (PP_4708) and RNase R (PP_4880). The presence of both exonucleases and multiple helicases is consistent with condition-dependent degradosome remodeling seen elsewhere in the genus, where cold-adapted *Pseudomonas* substitutes RNase R + RhlE for PNPase + RhlB.

The strongest, most directly transferable evidence comes from the same genus. Systematic mapping of the *P. aeruginosa* RNase E C-terminal scaffolding domain identifies short linear motifs (SLiMs) that recruit **PNPase and RhlB as the core partners** — establishing the *Pseudomonas* degradosome architecture that KT2440 almost certainly shares. However, mechanistic details diverge sharply from the *E. coli* paradigm: *Pseudomonas* RhlB belongs to a

distinct "**Type II**" **helicase clade** with an N-terminal intrinsically disordered region driving phase separation, and RNase E *antagonizes* rather than stimulates it. A sequence-similarity analysis quantified this: KT2440 degradosome proteins are roughly an order of magnitude more similar to their *P. aeruginosa* orthologs than to *E. coli* (e.g., RhlB ~52% vs. ~4% 5-mer Jaccard similarity). **Species transfer from *P. aeruginosa* is strong; transfer from *E. coli* is weak and must be flagged in curation.** Only Hfq (PP_4894) has direct functional characterization in KT2440 itself (Crc–Hfq translational repression).

For curation, the candidate list should be **trimmed of KEGG map co-listings and over-propagated annotations**: *recQ* (a DNA helicase), *rho* (transcription termination), *dnaK* and *groEL* (general chaperones, primary bucket ppu04156), and *ppk/ppkB* (polyphosphate kinases) are not core module members. Enolase (*eno*, PP_1612) is a bona fide *E. coli* degradosome subunit but is **most likely NOT a core Pseudomonas degradosome component** — its inclusion reflects *E. coli*-templated annotation, and the *P. aeruginosa* CTD interactome does not report an enolase-binding microdomain. The genes *rne*, *rhlB*, *pnp*, *rnr*, *rhIE*, and *rhIE-I* should be promoted to full `fetch-gene` review.

2. Target-Organism Pathway Definition

What the pathway includes

The module describes **RNase E-initiated messenger-RNA and stable-RNA turnover** — the ordered, mostly cytoplasmic process by which:

1. **Initiation / 5'-end sensing:** RNA pyrophosphohydrolase (RppH) converts protective 5'-triphosphate ends to 5'-monophosphate, the preferred substrate for RNase E; RNase E then makes internal endonucleolytic cuts.
2. **Scaffolding:** The RNase E C-terminal domain (CTD) nucleates a multi-enzyme "RNA degradosome" via short linear motifs.
3. **Unwinding:** A recruited DEAD-box RNA helicase (RhlB and/or RhlE) uses ATP to melt secondary structure that would otherwise block exonucleases.
4. **Processive decay:** A 3'→5' exoribonuclease (PNPase and/or RNase R) degrades the cleavage products to short oligonucleotides/mononucleotides.
5. **3'-end tagging (accessory):** Poly(A) polymerase PcnB adds 3' A-tails that stimulate exonucleolytic attack.

6. **Riboregulation (accessory)**: Hfq chaperones small RNAs and modulates target accessibility.

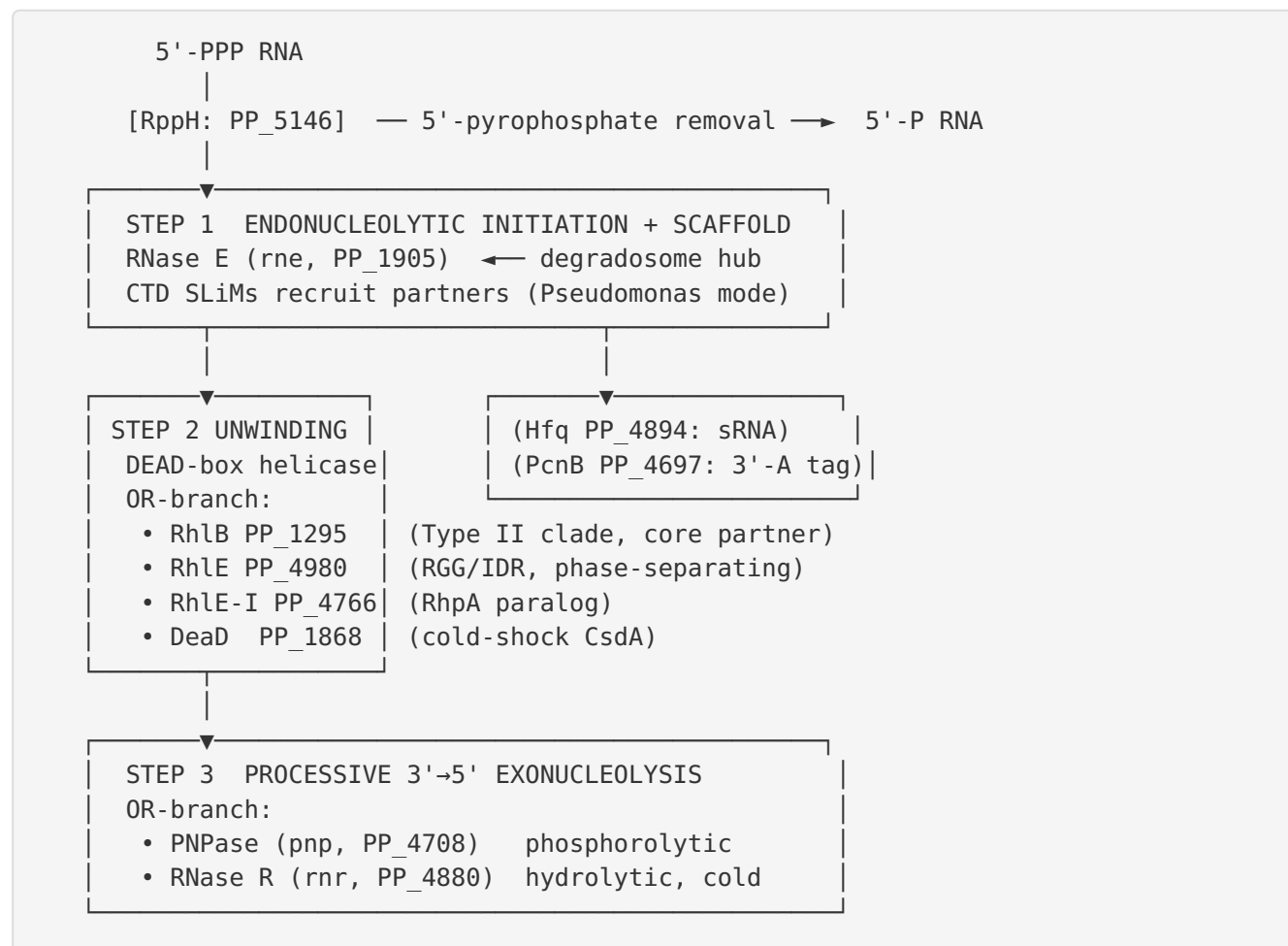
Neighboring pathways to keep separate

- **Transcription termination** (Rho): a distinct process; Rho terminates transcription and is not part of the decay machinery.
- **DNA repair/recombination** (RecQ): a DNA helicase, unrelated to RNA decay.
- **Protein folding / chaperones** (ppu04156: DnaK, GroEL): peripheral to degradosome assembly at best, not catalytic decay steps.
- **Polyphosphate metabolism** (Ppk, PpkB): energy/phosphate storage, not RNA decay.
- **Glycolysis / central carbon metabolism** (Enolase): a glycolytic enzyme whose *E. coli* degradosome moonlighting does not clearly transfer to *Pseudomonas*.

Alternate names and database definitions

- KEGG `ppu03018` = "RNA degradation" (organism-specific instance of the reference map `03018`).
 - The physical complex is the "**RNA degradosome**"; the *Pseudomonas* version is sometimes called the "cold-shock degradosome" when RNase R + RhlE are incorporated.
 - RNase E = EC 3.1.26.12; PNPase = EC 2.7.7.8; RNase R = EC 3.1.13.1; DEAD-box helicases = EC 3.6.4.13.
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3. Expected Step Model



Branch logic: Steps 2 and 3 are OR-steps. The module is satisfied if *at least one* helicase and *at least one* exonuclease are present. In KT2440 **all branches are encoded**, which is stronger than minimal satisfiability and consistent with condition-dependent degradosome composition.

4. Candidate Genes and Evidence

High-confidence core module genes

Gene	Locus	UniProt	Module step	Confidence	Evidence type
<i>rne</i> (RNase E)	PP_1905	Q88LM4	Step 1 scaffold + endonuclease	High	Genus (<i>P. aeruginosa</i>) direct + homology
<i>rhlB</i>	PP_1295	Q88NB7	Step 2 helicase (core partner)	High	Genus direct + sequence orthology
<i>pnp</i> (PNPase)	PP_4708	Q88DW0	Step 3 exonuclease (core partner)	High	Genus direct + homology
<i>rnr</i> (RNase R)	PP_4880	Q88DE6	Step 3 exonuclease (cold variant)	High	Genus (<i>P. syringae</i>) direct
<i>rhlE</i>	PP_4980	Q88D48	Step 2 helicase (variant)	High	Genus + sequence (IDR)
<i>rhlE-I</i> / RhpA	PP_4766	Q88DQ7	Step 2 helicase (paralog)	Medium	Sequence paralog resolution
<i>hfq</i>	PP_4894	Q88DD3	Accessory riboregulator	High	Direct KT2440 experiment
<i>rppH</i>	PP_5146	Q88CN4	Accessory 5'-trigger	Medium	Homology (<i>E. coli</i> mechanism)
<i>pcnB</i> (PAP I)	PP_4697	Q88DX1	Accessory 3'-tagging	Medium	Homology (<i>E. coli</i> mechanism)
<i>deaD</i> / CsdA	PP_1868	Q88LR1	Step 2 helicase (cold-shock)	Medium	Sequence paralog + homology

Gene-by-gene assessment

RNase E — *rne* (PP_1905). The conserved catalytic and scaffolding hub of the module. In *P. aeruginosa*, systematic CTD interactome mapping identified the SLiMs required for membrane attachment, RNA binding, complex clustering, and **direct binding to the core partners PNPase and RhlB** (PMID: 40096066). Cryo-EM/biochemistry showed the PNPase–RNase E

association is maintained across *E. coli*, *Salmonella*, and *P. aeruginosa* but through lineage-specific recognition modes (PMID: 41036625). KT2440 encodes a full-length RNase E (~1091 aa) with an intact C-terminal scaffold, so the scaffolding step transfers strongly from the genus. **Caveat:** the recognition interface is genus-specific — do not transfer *E. coli* CTD-microdomain annotations verbatim.

RhlB — *rhlB* (PP_1295). A core degradosome helicase in *Pseudomonas*. It belongs to a **distinct "Type II" clade** exemplified by *P. aeruginosa* RhlB, containing an N-terminal intrinsically disordered region that drives RNA-dependent liquid–liquid phase separation; *P. aeruginosa* RNase E binds it through an interface **distinct from the *E. coli* model** and **antagonizes** rather than stimulates its condensation, controlling growth at low temperature (PMID: 42581758). Sequence analysis confirms KT2440 RhlB (Q88NB7, 398 aa) is a near-identical ortholog of the *P. aeruginosa* Type II reference (Q9HXE5, 397 aa) with a compact architecture (~15 aa upstream of the DEAD-box Walker A GTGKT motif) like *E. coli* RhlB. **Caveat:** annotation should note the Type II regulatory divergence from *E. coli*.

PNPase — *pnp* (PP_4708). The phosphorolytic 3'→5' exonuclease and a core degradosome partner in *Pseudomonas* (PMID: 40096066; PMID: 41036625). High confidence for Step 3.

RNase R — *rnr* (PP_4880). The hydrolytic 3'→5' exonuclease alternative. In the psychrotroph *P. syringae* Lz4W, the degradosome contains **RNase R instead of PNPase**, together with RNase E and the DEAD-box helicase RhlE (PMID: 15705581); RNase R interacts directly with RNase E and is essential for growth at 4 °C, with a novel role in rRNA 3'-end processing (PMID: 17405875). This establishes the exonuclease-variant OR-branch as genuine within the genus. High confidence.

RhlE — *rhlE* (PP_4980). The RhlE-family helicase and the demonstrated cold-degradosome partner in *P. syringae* (PMID: 15705581). Sequence analysis shows KT2440 RhlE (Q88D48, 626 aa) carries a ~249-aa C-terminal extension that is strongly disorder-promoting (fraction A/R/G/Q/S/E/K/P = 0.76) and RGG/RG-rich (...GGGEKRPPRANNGGGARRDGGGGRGPARD...) — a classic phase-separation-prone IDR. **This is the phase-separating helicase in KT2440**, distinct from the compact RhlB.

RhlE-I / RhpA — *rhlE-I* (PP_4766). A second RhlE-family paralog (Q88DQ7, 443 aa) with a ~66-aa Lys/Arg-rich C-tail (disorder-promoting fraction 0.73). Resolved as a distinct paralog by sequence. Medium confidence; paralog ambiguity flagged.

DeaD / CsdA — *deaD* (PP_1868). Cold-shock DEAD-box helicase (Q88LR1, 559 aa, ~177-aa C-tail, disorder fraction 0.62). A fourth distinct paralog; likely a condition-specific helicase rather than the constitutive core partner. Medium confidence.

Hfq — *hfq* (PP_4894). The **only candidate with direct KT2440 evidence**: in *P. putida* KT2440, carbon catabolite repression is mediated by the **Crc–Hfq complex** binding the 5' region of target mRNAs to inhibit translation (PMID: 37348756). This directly validates Hfq's post-transcriptional-regulatory role in the target strain.

RppH — *rppH* (PP_5146). Accessory 5'-end decay trigger. In *E. coli*, RppH (formerly NudH/YgdP) removes pyrophosphate to convert 5'-triphosphate to 5'-monophosphate, initiating 5'-end-dependent decay via RNase E (PMID: 18202662). Role in KT2440 is homology-inferred (medium confidence).

PcnB (PAP I) — *pcnB* (PP_4697). Accessory 3'-tagging. In *E. coli*, 3' adenylation directly affects digestion by PNPase, and 3' adenylation and 5' phosphorylation act cooperatively (PMID: 7533264). Role in KT2440 is homology-inferred (medium confidence).

5. Gaps, Ambiguities, and Likely Over-Annotations

Likely KEGG co-listings / over-propagated annotations (recommend removal from module)

The KEGG RNA-degradation map co-lists helper/regulatory nodes that are **not** part of the RNase E endo–helicase–exo core:

Gene	Locus	Actual function	Why excluded
<i>recQ</i>	PP_4516	DNA helicase (EC 5.6.2.4)	DNA repair/recombination; no RNA-decay role
<i>rho</i>	PP_5214	Transcription termination factor Rho	Transcription, not decay
<i>dnaK</i>	PP_4727	Chaperone HSP70	General chaperone (primary bucket ppu04156)
<i>groEL</i>	PP_1361	Chaperonin GroEL	General chaperone (primary bucket ppu04156)
<i>ppk</i>	PP_5217	Polyphosphate kinase (EC 2.7.4.1)	PolyP metabolism
<i>ppkB</i>	PP_0712	ADP/GDP-polyphosphate phosphotransferase	PolyP metabolism

These are peripheral degradosome-assembly/cold-shock helpers at best (DnaK/GroEL) or wholly unrelated (RecQ, Rho, Ppk/PpkB). They should not count toward module satisfiability.

Enolase — the key over-annotation call

Enolase (*eno*, PP_1612) is a genuine and well-documented degradosome subunit in *E. coli*/Enterobacteriaceae, where it is recruited via a short, poorly conserved CTD microdomain. However, its membership in the **Pseudomonas** degradosome is **not established**. The systematic *P. aeruginosa* RNase E CTD interactome reported binding microdomains for **PNPase and RhlB only — no enolase-binding microdomain** (PMID: 40096066). The only *P. aeruginosa* mention of an "enolase dimer" appears in a generic degradosome description within a structural RraA paper (PMID: 21063756), **not** a demonstration of direct binding. KT2440 *eno* carries a canonical glycolytic annotation (EC 4.2.1.11). **Recommendation: treat enolase as glycolytic, not a core Pseudomonas degradosome subunit; its ppu03018 listing reflects E. coli-templated annotation.**

Paralog ambiguity

Four distinct DEAD-box helicase paralogs are resolved by sequence: RhlB (compact core partner, 398 aa), RhlE (RGG/IDR, phase-separating, 626 aa), RhlE-I/RhpA (short K/R-rich tail, 443 aa), and DeaD/CsdA (cold-shock, 559 aa). Curators should ensure each locus retains its specific paralog identity and not be collapsed under a generic "EC 3.6.4.13 DEAD-box helicase" annotation. RhlE vs. RhlB identity is the most important distinction because they play mechanistically different roles (RhlB = constitutive core; RhlE = cold/structured-substrate variant).

Species-transfer strength (quantified)

A conservative 5-mer Jaccard similarity proxy showed KT2440 degradosome proteins are far closer to *P. aeruginosa* (same genus) than to *E. coli*:

Protein	vs. <i>E. coli</i>	vs. <i>P. aeruginosa</i>
RhlB	3.8%	52.5% (containment 68.8%)
PNPase	10.6%	41.4% (containment 58.6%)
Hfq	21.6%	44.1% (containment 59.8%)
RNase E	9.5%	(not retrieved; expected high)
RNase R	5.1%	(not retrieved; expected high)
RhlE	9.4%	(not retrieved; expected high)

Curation rule of thumb: transfer functional detail from *P. aeruginosa*/genus with high confidence; transfer from *E. coli* only for broad enzyme identity, and flag mechanistic/regulatory details (especially RhlB regulation and enolase membership) as *E. coli*-specific.

6. Mechanistic Model / Interpretation

Synthesizing the findings, KT2440 possesses a **complete, genus-typical, and condition-flexible RNase E degradosome**. RNase E serves as the membrane-associated scaffold whose C-terminal disordered domain presents SLiMs recruiting the core partners PNPase and RhlB ([PMID: 40096066](#)). Unlike the textbook *E. coli* complex, the *Pseudomonas* complex (a) uses genus-specific recognition interfaces for PNPase ([PMID: 41036625](#)), (b) recruits a **Type II RhlB** whose activity is *antagonized* — not stimulated — by RNase E through phase-separation control ([PMID: 42581758](#)), and (c) **likely does not incorporate enolase** as a core subunit ([PMID: 40096066](#)).

Under cold stress, the genus can remodel the complex to swap RNase R for PNPase and RhlE for RhlB ([PMID: 15705581](#), [PMID: 17405875](#)), and KT2440 encodes all the parts (RNase R PP_4880, RhlE PP_4980, DeaD PP_1868) to do so. Upstream, RppH primes transcripts by 5'-pyrophosphate removal ([PMID: 18202662](#)) and PcnB tags 3' ends to accelerate exonucleolysis ([PMID: 7533264](#)); Hfq, the only directly validated KT2440 component, integrates small-RNA and catabolite-repression signals ([PMID: 37348756](#)). The net picture is a module that is not merely "present" but **redundantly and adaptively provisioned** in this organism.

The degradosome remodeling logic can be summarized as:

Condition	Endonuclease	Helicase	Exonuclease
Standard (mesophilic)	RNase E (PP_1905)	RhlB (PP_1295)	PNPase (PP_4708)
Cold / structured substrates	RNase E (PP_1905)	RhlE (PP_4980) / DeaD (PP_1868)	RNase R (PP_4880)

7. Module and GO-Curation Recommendations

Module step	Status	Rationale
Step 1: RNase E cleavage / scaffold	covered	<i>rne</i> PP_1905; genus-direct scaffold evidence (P 40096066 P 41036625)
Step 2: RhlB-assisted unwinding	covered	<i>rhlB</i> PP_1295; Type II ortholog of <i>P. aeruginosa</i> (P 42581758)
Step 2: RhlE-assisted unwinding	covered	<i>rhlE</i> PP_4980 + genus precedent (P 15705581)
Step 3: PNPase phosphorolysis	covered	<i>pnp</i> PP_4708; core partner (P 40096066)
Step 3: RNase R hydrolysis	covered	<i>rnr</i> PP_4880; genus-direct (P 15705581 P 17405875)
Accessory: 5'-trigger (RppH)	candidate_uncertain	Homology only; no KT2440 data
Accessory: 3'-tagging (PcnB)	candidate_uncertain	Homology only; no KT2440 data
Accessory: Hfq riboregulation	covered	Direct KT2440 evidence (P 37348756)
Enolase as core subunit	not_expected_in_target_taxon	No <i>Pseudomonas</i> enolase-binding microdomain (P 40096066)
RecQ, Rho, DnaK, GroEL, Ppk, PpkB	not module members	KEGG co-listings / other buckets

Module-boundary note: The generic module models Steps 2 and 3 as OR-branches with a single selected implementation. KT2440 encodes **all** branches, which the module should treat as full satisfiability. The generic boundaries are broadly correct, but the **Pseudomonas instance should explicitly exclude enolase** from the degradosome core (unlike the *E. coli*-derived generic degradosome) and should annotate RhlB with the **Type II** regulatory divergence.

GO-curation: No new GO terms appear strictly required. Existing terms cover ribonuclease E activity (GO:0008995), RNA helicase activity (GO:0003724), polyribonucleotide nucleotidyltransferase activity (GO:0004654), and exoribonuclease activity. A curator note distinguishing "RNA degradosome" assembly (*Pseudomonas* mode) from the *E. coli* mode would be valuable, and a request to capture **RhlB Type II phase-separation antagonism by RNase E** as an annotation extension could be considered.

8. Genes to Promote to Full fetch-gene Review

Promote the following core module genes to full individual review, in priority order:

1. **rne (PP_1905)** — the module hub; verify CTD SLiM inventory and *Pseudomonas*-specific partner interfaces.
2. **rhlB (PP_1295)** — confirm Type II clade assignment and IDR annotation; reconcile with generic RhlB.
3. **pnp (PP_4708)** — core exonuclease; confirm degradosome membership.
4. **rnv (PP_4880)** — cold-variant exonuclease; confirm RNase E interaction potential in KT2440.
5. **rhlE (PP_4980)** — resolve RhlE vs. RhlB roles; annotate the RGG/IDR.
6. **rhlE-I / RhpA (PP_4766)** — paralog disambiguation.

Secondary (accessory) review: *hfq* (PP_4894, already validated), *rppH* (PP_5146), *pcnB* (PP_4697), *deaD* (PP_1868). Enolase (PP_1612) should be reviewed specifically to **downgrade** its degradosome annotation to glycolytic.

9. Limitations and Knowledge Gaps

- **Direct KT2440 evidence is thin.** Only Hfq has been experimentally characterized in the target strain. Nearly all mechanistic conclusions rely on transfer from *P. aeruginosa* or *P. syringae*. The transfer is well-justified by high sequence similarity within the genus, but no KT2440 degradosome has been directly isolated or its composition mass-spec-verified.
- **Sequence proxy caveats.** The 5-mer Jaccard analysis is a conservative, divergence-sensitive proxy, not a phylogenetic or structural alignment; *P. aeruginosa* orthologs for RNase E, RNase R, and RhlE were not retrieved by exact gene-name search, so their genus-transfer strength is inferred by trend rather than measured.
- **Enolase call is negative evidence.** The conclusion that enolase is not a core *Pseudomonas* degradosome subunit rests on its *absence* from a reported interactome, which is weaker than a direct binding-negative experiment.
- **Accessory step uncertainty.** RppH and PcnB roles are entirely homology-inferred; their actual contribution to decay in KT2440 is untested.
- **Condition dependence untested in KT2440.** Whether KT2440 actually swaps RNase R/RhlE into its degradosome under cold stress (as *P. syringae* does) is an extrapolation.

10. Proposed Follow-up Experiments / Actions

1. **Affinity-purify the KT2440 RNase E degradosome** (tagged *rne* PP_1905) under standard and cold-shock conditions; identify partners by LC-MS/MS. This would directly confirm PNPase/RhlB core membership, test RNase R/RhlE cold substitution, and settle the enolase question in the target strain.
2. **Map the KT2440 RNase E CTD SLiM inventory** in silico against the *P. aeruginosa* reference (

P 40096066

 to predict partner-binding microdomains and confirm absence of an enolase microdomain.
3. **Phenotype the KT2440 *rppH* and *pcnB* deletions** with 5'-end and 3'-tail sequencing to test their inferred accessory roles directly.
4. **Confirm RhlB Type II assignment** by phylogenetic placement and IDR prediction, and test RNase E antagonism of RhlB condensation in KT2440 (following

P 42581758

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5. **Curation actions:** trim RecQ, Rho, DnaK, GroEL, Ppk, PpkB from the module; downgrade enolase to glycolytic; promote *rne*, *rhlB*, *pnp*, *rnR*, *rhlE*, *rhlE-I* to full review; annotate all four DEAD-box loci with distinct paralog identities.

11. Evidence Base / Key References

PMID	Title (abbrev.)	How it supports the review
40096066	Critical functions of the RNase E scaffolding domain in <i>P. aeruginosa</i>	Establishes PNPase + RhlB as core degradosome partners; no enolase microdomain reported
42581758	RNase E resolves toxic condensates of Type II RhlB helicases	Defines Pseudomonas Type II RhlB and its E. coli-divergent regulation
41036625	Multi-dentate endo–exo interaction in the bacterial degradosome	RNase E–PNPase association conserved but genus-specific in Pseudomonas
15705581	RNase R interacts with RNase E + helicase in <i>P. syringae</i>	Supports RNase R and RhlE variant branches in Pseudomonas
17405875	RNase R essential at low temperature in <i>P. syringae</i>	Direct RNase R–RNase E degradosome interaction in the genus
37348756	Crc–Hfq system of <i>P. putida</i> KT2440	Direct KT2440 evidence for Hfq function
18202662	RppH triggers mRNA degradation by 5'-pyrophosphate removal	Establishes RppH's 5'-trigger role upstream of RNase E
7533264	RNA degradation regulated by 3' adenylation and 5' phosphorylation	Establishes PcnB stimulation of PNPase degradation
21063756	Crystal structure of hexameric RraA from <i>P. aeruginosa</i>	Context for Pseudomonas degradosome; enolase mention is generic, not direct binding
27447594 / 27834591	Phage Dip inhibits the <i>P. aeruginosa</i> degradosome	Confirms RNase E scaffold has conserved RNA-binding sites in Pseudomonas

Report prepared for manual module satisfiability and gene-annotation curation. All species-transfer judgments are stated explicitly; direct KT2440 evidence is limited to Hfq, with the remaining core mechanism transferred with high confidence from the same genus (P. aeruginosa, P. syringae) and only weakly from E. coli.

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